

## Work distribution for Assignment (Project)

| Section                                                                                                  | Task Distribution                                |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Abstract & Area of study & Keywords                                                                      | Member 1                                         |
| Chapter 1 – Problem Statement and Motivation                                                             | Member 2                                         |
| Chapter 1 – Scope, Objectives                                                                            | Member 3                                         |
| Chapter 1 – Contribution                                                                                 | Member 4                                         |
| Chapter 1 – Report Organization                                                                          | Member 4                                         |
| Chapter 2 – Literature review 1                                                                          | Member 1                                         |
| Chapter 2 – Literature review 2                                                                          | Member 2                                         |
| Chapter 2 – Literature review 3                                                                          | Member 3                                         |
| Chapter 2 – Literature review 4                                                                          | Member 4                                         |
| Chapter 2 Comparison of the 4 techniques                                                                 | Each student contributes here                    |
| Chapter 3 – System Design / Overview                                                                     | Member 1                                         |
| Chapter 3 – System Requirement<br>(Hardware and software to use)                                         | Member 2                                         |
| Chapter 3 – Implementation Issues and Challenges                                                         | Member 3                                         |
| Chapter 3 – Timeline & milestones for 13 weeks                                                           | Member 4                                         |
| Chapter 4 – Coding work 1 & results & evaluations                                                        | Member 1                                         |
| Chapter 4 – Coding work 2 & results & evaluations                                                        | Member 2                                         |
| Chapter 4 – Coding work 3 & results & evaluations                                                        | Member 3                                         |
| Chapter 4 – Coding work 4 & results & evaluations                                                        | Member 4                                         |
| Chapter 5 – Conclusion                                                                                   | One paragraph Member 1<br>One paragraph Member 2 |
| Chapter 5 - Recommendation                                                                               | One paragraph Member 3<br>One paragraph Member 4 |
| References (in IEEE style)                                                                               | Each student contributes here                    |
| Appendices (e.g. list of classified test images results/<br>list sample code/source code by each member) | Each student contributes here                    |

## UCCC2513 Project Work

### Traffic sign detection for autonomous vehicles

Continuing from the proposal, the task here is to complete a system to detect traffic signs reliably for autonomous vehicles. Students should work in groups of 4 members. Eighty-four (84) test images are provided in the directory “Traffic signs” (which can be found in the directory of “Inputs” at MS Teams) for testing purpose. Figure 1 below shows one example of the **test image** and the desirable detection result “Sign 60”.



Figure 1: One example of the test image and detection result

**To get enough images for training and validation**, one can go through the Internet or AI to download other datasets of traffic signs.

#### 1. **OBJECTIVE**

Design and implement using the learnt image processing techniques an automatic traffic sign detector using OpenCV to segment and identify the sign correctly from the 84 provided test input images. The sub-tasks include

- a) Segment each sign out of an image.
- b) Identify each segmented sign as sign 1, sign 2, ....., sign 99.
- c) The operation should run in real time and take less than 2 seconds on any of the lab machines for each image.

#### 2. **REPORT**

You need to write a report for your work. The report should follow the template file “FYP2\_Report\_Template\_CS-(New)” from the FICT FYP web site. It should include the followings:

- a. **W<sup>5</sup>H**: This represents the terms What, Why, When, Who, Where and How with respect to the assignment title in writing report. Usually, we pay more attention to what, why and how. In this report, you need to provide and explain what the system design is (i.e. the system block diagram or

- flowchart), why this design and how each of the details work together to form the final system.
- Experimental result and error analysis.** In addition, under what conditions will your system fail and why?
  - The source code listings.
  - Choices of thresholding and tolerance parameters.** Give reasons to justify your choices.

Other small but important details that you should know are:

- Excluding the code and intermediate results shown in the appendices, the written report should **not exceed 80 pages** with double spacing and font sizes 10 to 12 using font Times New Roman. The number “80” is obtained from the counting of pages from the file “FYP2\_Guidelines”. Usually, a **60 pages** mini-project report is expected since it is not a FYP report. A long report might imply that you are not able to do **summarization** and hence producing a **second class report** (should be penalized).
- Report will be graded on **clarity** and **authenticity**.
- Each report section and source code section should be **tagged with name** of the group member who produces the writing and / or the implementation.
- University regulation prohibits **plagiarism**. If caught, you will end up with an “F” grade.
- Late project **will not be** accepted after the due date under any circumstances.
- Each team member is required to choose and develop one specific module as listed below:

|               |                                                                                                                                                                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Task 1</b> | Identify image features to aid the identification work. This task needs <b>two students</b> . Each student should describe his/her contribution separately without consulting the other. Report the results of recognition rate according to different sets of features. |
| <b>Task 2</b> | Identify a few classifiers to do the recognition. This task needs <b>two students</b> . Each student should describe his/her contribution separately. Report the results of recognition rates according to different classifiers.                                        |

- Google Form Submission Link:  
<https://docs.google.com/forms/d/e/1FAIpQLSenPprivS7SjOv7l5e4nnr4YU7d6oQ1C947lhXUk-CmyaCwPg/viewform?usp=dialog>
- Submission deadline: **7<sup>th</sup> September 2026 by 12pm**

### 3. **PRESENTATION & DEMONSTRATION**

Besides the report, you are required to provide a short video of presentation of your work and demonstration running your code and displaying the results. The demo should read the names of input image files from a file named “inputFiles.txt”. Each name is on a separate line in the file.

Presentation Requirements:

1. The total presentation and Q&A session for each group shall not exceed **30 minutes**.
2. The recorded short video (proposal presentation, code demonstration, and preliminary results) shall be **a maximum of 25 minutes** in duration + followed by a **physical Q&A session of 5 minutes**.
3. The video quality must be clear and suitable for viewing.
4. Audio quality must be clear, audible, and free from significant background noise.
5. All group members are required to present their respective components in the short video presentation and participate in the physical Q&A session.

### 4. **MARKING**

| Items | Contents                    | Marks                                    |
|-------|-----------------------------|------------------------------------------|
| 1     | Project report (teamwork)   | <b>30</b> (report 20<br>Presentation 10) |
| 2     | Project report (individual) | <b>20</b>                                |

Item 1 is based on whether a clear and easily understood presentation & demo have been presented.

The markings of items 1 and 2 are based on the following criteria where parts (a), (c) and (d) will award marks according to the FYP2 rubrics from the FICT FYP web site. In total, 100 marks will be awarded below for items 1 or 2 but each will be scaled for the final **30** or **20** marks as shown in the above table.

- a) A well written report, **20** marks.
  - i. Follow the FYP2 report writing format to explain what the system design (i.e. the system block diagram or flowchart) is, why this design and how each of the details work together to form the final system. (13 marks)
  - ii. Experimental result and error analysis. (4 marks)
  - iii. Choices of thresholding and tolerance parameters. Give reasons to justify your choices. (2 marks)
  - iv. Report is within 80 pages long. (1 marks)
- b) Your work can run within 2 seconds with any of the lab machines. (5 marks)
- c) For each sign, correct identification is worth 1 mark. (**70** marks)

- d) The demo can go beyond the provided 84 test images. (5 marks)

5. **HINTS**

One can find numerous pieces of demo and code using OpenCV from the Internet to segment traffic sign. You can refer / reuse them for your code. **Cite the sources** in your report if you reuse or adapt their code. In addition, you need to test your code for all the test images.